

Sensitivity study of meteorological parameters on mineral aerosol mobilization, transport, and distribution

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[1] Predicting mineral aerosol distributions is a difficult task due to the episodic nature of the sources and transport. Here we show comparisons between a 22-year simulation of mineral aerosols and satellite and in situ observations. Our results suggest that the model does a good job of predicting atmospheric mineral aerosol distributions, with some discrepancies. In addition, there are differences between our model results and previously published results [e.g., *Ginoux et al.*, 2001]. We conduct several tests of the sensitivity of mineral aerosol simulations to the meteorological data sets and mobilization parameterizations in order to understand the differences. Comparisons between model simulations using National Center for Atmospheric Research/National Center for Environmental Prediction (NCEP/NCAR) and National Aeronautics and Space Administration Data Assimilation Office (NASA DAO) reanalysis data sets show that the model results with the two data sets are fairly consistent but with some important differences. The sensitivity analysis shows that differences between simulated dust near Australia are likely due to differences in both source parameterization and surface winds. Differences over East Asia are dominated by differences in meteorology. The sensitivity analysis also shows that we cannot tell from comparisons with observations whether the cultivation source is active nor eliminate it because of the large uncertainty in meteorology and source parameterization. **INDEX TERMS:** 0305 Atmospheric Composition and Structure: Aerosols and particles (0345, 4801); 0315 Atmospheric Composition and Structure: Biosphere/atmosphere interactions; 0322 Atmospheric Composition and Structure: Constituent sources and sinks; 0330 Atmospheric Composition and Structure: Geochemical cycles

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1. Introduction

[2] Mineral aerosols are suggested to play an important role in climate forcing by altering the radiation balance in the atmosphere [e.g., *Miller and Tegen*, 1999]. In addition, mineral aerosols can provide surface area for heterogeneous reactions in the atmosphere and significantly affect the cycles of atmospheric species [e.g., *Dickerson et al.*, 1997; *Dentener et al.*, 1996]. Finally, mineral aerosol deposition impacts nutrient cycles in ocean and terrestrial ecosystems, and thereby could play an important role in modulating the global carbon cycle [e.g., *Martin*, 1990; *Archer et al.*, 2000; *Watson et al.*, 2000; *Chadwick et al.*, 1999]. There are major uncertainties in modeling of dust aerosol, due to the sparseness of the observational data, and gaps in our understanding of the physical and chemical properties of mineral aerosol. Mineral aerosol entrainment into the atmosphere (called mobilization) is sensitive

to a wide range of factors including the composition of the soils, the soil moisture content, surface condition, and wind speed and may be modulated by human activities and land degradation. Because detailed soil information and source information is not available, there are differences in the source parameterizations used in different models. In addition, while several models use reanalysis winds [e.g., *Schulz et al.*, 1996; *Ginoux et al.*, 2001; *Tegen et al.*, 2002], in areas with few meteorological observations such as desert areas, it is likely that important meteorological parameters such as surface wind speeds are not well observed, but rather sensitive to forecast center model.

[3] In this study we extend the previous studies by using a slightly different mobilization scheme, different transport model, and a different set of assimilated winds, to look at the period from 1979 to 2000. We compare the climatology of mineral aerosols produced from this 22-year simulation to satellite and in situ data sets to show that the model predictions are realistic. We also contrast these results with similar global studies conducted by *Ginoux et al.* [2001]. In order to understand the differences between the *Ginoux et al.* [2001] and our results, we conduct several sensitivity studies isolating meteorology and source parameterizations. *Ginoux et al.* [2001] used the Georgia Tech/Goddard Global Ozone Chemistry Aerosol Radiation and Transport

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(GOCART) model, driven by the Goddard Earth Observing System Data Assimilation System (GEOS DAO). We used the MATCH chemistry transport model [Rasch et al., 1997; Mahowald et al., 1997] coupled with Desert Entrainment And Deposition model [Zender et al., 2003] driven by National Center for Environmental Prediction/National Center for Atmospheric Research (NCEP) reanalysis data. Here we explore how changes in the meteorological data sets and the source parameterization change our results, and especially focus on understanding differences between Ginoux et al. [2001] and our results.

[4] In Section 2 we describe the model simulations conducted in this paper. In Section 3 we show the results of a 22-year model simulation and compare to available observations. Section 4 shows results of meteorological and source change sensitivity studies. In Section 5 we summarize the results of this paper.

2. Model Description

[5] For this study we use Model of Atmospheric Transport and Chemistry (MATCH), [Rasch et al., 1997; Mahowald et al., 1997], coupled with Desert Entrainment And Deposition (DEAD) model [Zender et al., 2003; Mahowald et al., 2002]. This mobilization scheme is based on the wind tunnel and in situ studies of Iversen and White [1982], Marticorena and Bergametti [1995], Gillette et al. [1997], and Fecan et al. [1999]. It is similar to those used by Tegen and Fung [1994], Mahowald et al. [1999], Guelle et al. [2000], Ginoux et al. [2001], and Tegen et al. [2002] in that it is based on a wind threshold velocity and has a wind speed cubed relationship for dust mobilization, but the details of the mobilization are slightly different in each case. The DEAD module mobilization scheme [Zender et al., 2003] is also based on the friction velocity (wind stress at the surface) not the wind velocity on the surface level in the model, consistent with the original derivations [e.g., Gillette et al., 1997]. The wind stress is a function of the wind speed, roughness length and the atmospheric stability at the surface. The surface roughness length in dust producing regions is set to the globally uniform value of 100 μm , a value typical of erodible soil beds [Gillette et al., 1997] than the large-scale roughness lengths for bare ground (~ 5 cm) used in general circulation models [e.g., Bonan, 1996]. The mass flux of saltating particles depends on the excess of the wind friction speed over the threshold wind friction velocity for saltation. The saltating particle flux was calculated by White's [1979] formulation.

$$F_h = c \frac{\rho_a}{g} U^{*3} \left(1 - \frac{U_t^*}{U^*}\right) \left(1 + \frac{U_t^*}{U^*}\right)^2 \quad (1)$$

where c is constant, ρ_a is the atmospheric density, U^* is the wind friction speed, g is acceleration of gravity, and U_t^* is the threshold wind friction speed. The threshold velocity is calculated by a semiempirical parameterization [Iversen and White, 1982].

for $0.03 < \text{Re}_{*t} < 10$

$$U_t^* = \frac{0.129K}{(1.928\text{Re}_{*t}^{0.092} - 1)^{0.5}} \quad (2)$$

for $\text{Re}_{*t} > 10$

$$U_t^* = 0.129K [1 - 0.0858 \exp(-0.0617(\text{Re}_{*t} - 10))] \quad (3)$$

$$\text{with } k = \left(\frac{\rho_p g D_p}{\rho_a}\right)^{0.5} \left(1 + \frac{0.006}{\rho_p g D_p^{2.5}}\right)^{0.5},$$

where ρ_p is particle density, ρ_a is atmospheric density, D_p is particle diameter, g is acceleration of gravity, and Re_{*t} is threshold friction Reynolds number. Based on ratio of vertical to horizontal flux experimentally determined, Gillette and Passi [1988] have suggested considering it as a linear function of the wind friction velocity. The relation between the vertical and horizontal flux for $0\% < (\% \text{ clay}) < 20\%$ is

$$\log(F_v/F_h) = 0.134(\% \text{ clay}) - 6 \quad (4)$$

We used globally uniform value of $\% \text{ clay} = 20\%$.

[6] Soil moisture is accounted for by increasing threshold friction velocity following Fecan et al. [1999]. The fraction of bare soil exposed in a gridcell and suitable for mobilization is the maximally overlapped product of the fractions of dry ground, non-snow-covered ground, and nonvegetated ground [Ginoux et al., 2001; Zender et al., 2003]. Snow cover is derived from the liquid water equivalent snow depth provided by the analyses. The mobilization scheme is sensitive to wind velocity, atmospheric stability and soil wetness. Because accurate soil characteristics are not available at sufficient resolution globally, we assume that soils are replete with particles optimally sized to initiate saltation, and use a factor to describe what fraction of each grid box consists of easily erodible soils. In this study, we follow Ginoux et al. [2001], Zender et al. [2003], and Mahowald et al. [2002], and assume that all topographic lows with little vegetation and low soil moisture are dust sources, using the time independent source area from Ginoux et al. [2001], which includes only nonvegetated low-lying regions (see Figure 1a). Thus the mobilization (or emission of dust into the atmosphere) is derived from multiplying the source area [from Ginoux et al., 2001] by the mobilization calculated using the DEAD dust module [Zender et al., 2003]. The fractions of the clay (dust particles smaller than $1 \mu\text{m}$) and silt sizes are different for different soil types at each location. Due to the uncertainties in the available soil texture data, following Tegen and Fung [1994] and Ginoux et al. [2001], we chose a globally constant particle size distribution; the fractions are 0.1 for the class $0.1 - 1 \mu\text{m}$, and .3 for the classes $1 - 2.5$, $2.5 - 5.0$, and $5.0 - 10 \mu\text{m}$, where the sizes are diameters of the particle. Within each bin we assume log normal distributions in aerosol sizes [Zender et al., 2003].

[7] Both dry deposition and wet deposition are included as loss processes for the mineral aerosols. Dry depositional processes are simulated following Seinfeld and Pandis [1996] and include turbulent deposition and gravitational settling, with the latter dominating for large particles. We chose a wet scavenging efficiency of 750 kg/kg for all aerosol sizes and for convective and stratiform precipitation [Tegen and Fung, 1994], which lies with the range of measurements for clay-size particles, and results in a wet deposition lifetime of the dust of approximately 12 days.

[8] Transport is calculated using the MATCH off-line chemical transport model [Rasch et al., 1997; Mahowald et al., 1997], which has been shown to simulate wet and dry convective mixing, in addition to large-scale precipitation

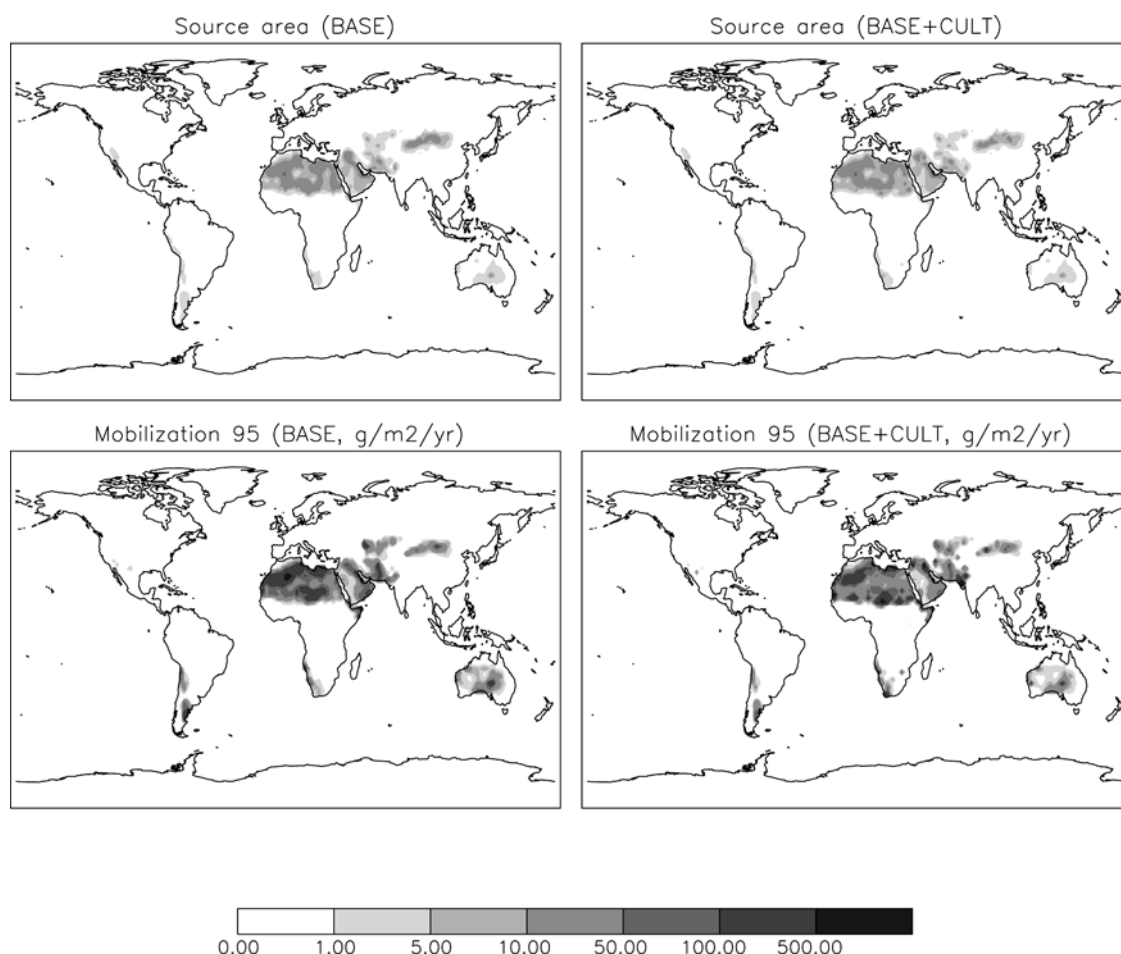


Figure 1. Dust mobilization for control run (BASE) and Disturbed source (Cult) run, in units of $gm^{-2}yr^{-1}$.

processes, well using the NCEP reanalysis [Mahowald *et al.*, 1997]. The horizontal resolution of the model is T62 ($\sim 1.9 \times 1.9$ degree), and 28 vertical levels from surface to 10 mb (the same as the resolution of the NCEP reanalysis made available at NCAR).

[9] We also compare against model results from Ginoux *et al.* [2001], and so we briefly describe the models used in those studies. Ginoux *et al.* [2001] use the Georgia Tech/Goddard Global Ozone Chemistry Aerosol Radiation and Transport (GOCART) model, driven by the GEOS DAS for the years 1987–1990 and 1996. The second source scheme comes from the GOCART model [Ginoux *et al.*, 2001] and it uses the same source area as in our BASE case. The dust mobilization based on empirical formulation by Gillette and Passi [1988].

$$F_p = CSs_p U_{10m}^2 (U_{10m} - U_t) \quad \text{if } U_{10m} > U_t$$

$$F_p = 0 \quad \text{otherwise}$$
(5)

where C is a dimensional factor, S is source function, U_{10m} is the horizontal wind speed at 10 m, and U_t is the threshold velocity, s_p and is the fraction of each size class. Dust uplifting requires the knowledge of the surface wind and the threshold velocity of wind erosion. The threshold velocity depends on the particle size and soil moisture. The modified

Belly's [1964] relationship was used to calculate threshold velocity. The threshold velocity was calculated by

$$U_t = A \sqrt{\frac{(\rho_p - \rho_a) g \phi_p}{\rho_a}} (1.2 + 0.2 \log_{10} w) \quad \text{if } w < 0.5$$

$$U_t = \infty \quad \text{otherwise}$$
(6)

where $A = 6.5$ is a dimensionless parameter, w is the surface wetness, ϕ_p is the particle diameter, g is the acceleration of gravity, ρ_p and ρ_a are the particle and air density, respectively [Ginoux *et al.*, 2001].

[10] The GOCART model uses 7 bins, and the dry deposition includes the turbulent transfer and gravitational settling. The turbulent transfer of the particles at the surface is calculated as a first-order process using a deposition velocity, which is assumed to be equivalent to the exchange velocity for heat and moisture at the surface. The settling velocity for the particle is determined by the Stokes law [Ginoux *et al.*, 2001]. Wet deposition includes rainout and washout in large-scale precipitation and in deep convective cloud updraft [Chin *et al.*, 2000].

2.1. Description of Sensitivity Studies

[11] In order to better understand the differences between the climatology presented here and that presented by Ginoux

Table 1. Sensitivity Study Descriptions

	BASE	GDD	BASE-DD	GDN	BASE+CULT
Source Area	Topographic lows [<i>Ginoux et al.</i> , 2001]	Topographic lows [<i>Ginoux et al.</i> , 2001]	Topographic lows [<i>Ginoux et al.</i> , 2001]	Topographic lows [<i>Ginoux et al.</i> , 2001]	Topographic lows + cultivation in deserts
Mobi. Scheme	DEAD, [<i>Zender et al.</i> , 2003]	GOCART, [<i>Ginoux et al.</i> , 2001]	DEAD, [<i>Zender et al.</i> , 2003]	GOCART, [<i>Ginoux et al.</i> , 2001]	DEAD [<i>Zender et al.</i> , 2003]
Data set for mobil. Calc.	NCEP	DAO	DAO	DAO	NCEP
Data set for transport	NCEP	DAO	DAO	NCEP	NCEP

et al. [2001], we conduct several sensitivity studies. First, it should be noted that all the sensitivity runs in this paper are based on same transport, same dry and wet removal processes [Mahowald *et al.*, 2002] and, for most simulations, the same source area following *Ginoux et al.* [2001]. The differences are mobilization schemes and meteorological input fields. Our BASE mobilization scheme is that described above [Zender *et al.*, 2003] and is based on threshold wind friction velocity and has a wind friction speed cubed relationship for dust mobilization. The scheme is based on the friction velocity (wind stress at the surface) not the wind velocity on the surface level in the model [e.g., Gillette *et al.*, 1997; Mahowald *et al.*, 2002; Zender *et al.*, 2003].

[12] The second source scheme comes from the GOCART model [Ginoux *et al.*, 2001] and it uses the same source area as in our BASE case. The dust mobilization based on empirical formulation by Gillette and Passi [1988] as described above. In *Ginoux et al.* [2001], the wind threshold velocity is set to 1.5 ms^{-1} for the smallest class and 3 ms^{-1} for the largest class. So this mobilization scheme has increasing mobilization with decreasing particle size. The dust flux is approximated as the wind speed (at 10 m) cubed by source function when wind at 10 meter larger than wind threshold velocity, otherwise dust flux is zero [Ginoux *et al.*, 2001]. We use the same bin sizes as in the BASE case for all sensitivity analyses, so that our mobilization will not be exactly the same as in *Ginoux et al.* [2001].

[13] We obtained NASA Data Assimilation Office (DAO) meteorological reanalysis for 1995 and interpolated these wind fields onto the same horizontal grid as the NCEP reanalyses (T62), thereby slightly increasing the resolution from 2.5×2.5 to 1.8×1.8 degrees by degrees. We used the DAO data sets on the original vertical coordinate, which has 20 levels, in contrast to the NCEP reanalysis 28 levels. Since it has not previously been shown that MATCH works well with DAO, we show a brief analysis to suggest that the subgrid scale mixing derived from NASA/DAO winds are robust in Section 4.1.

[14] Five sensitivity studies or cases were conducted for 1995: First, a base case or control run BASE (1995 from the 22-year climatology presented here). The mobilization, transport, and dry and wet deposition have been described above. The second case is GDD (GOCART mobilization scheme, mobilization calculated by DAO wind, with DAO transport). In this sensitivity, we turned off BASE case mobilization scheme, and implemented GOCART's mobilization [Ginoux *et al.*, 2001]. The transport, dry and wet remove processes of GDD are same as BASE case. Dust mobilization, transport and dry and wet removal processes were calculated using the DAO meteorological data set. The third case is BASE-DD (BASE model, but with mobilization calculated by DAO wind, and with DAO transport). All

schemes are same as BASE, which include mobilization, transport, dry and wet removal processes, but using the DAO data. The fourth case is GDN (GOCART mobilization scheme, mobilization calculated by DAO winds, with NCEP-derived transport). GDN's mobilization scheme and mobilization calculation are same as GDD, but the transport by NCEP data. The final case is BASE+CULT (BASE+ cultivation source, mobilization calculated by NCEP wind, and NCEP transport). The cultivation source is described in the following section. Table 1 gives a brief description of the sensitivity analyses.

2.2. Cultivation Source

[15] There has been considerable interest in understanding the role of the disturbed soils as dust sources. But efforts to model the impacts of land use on dust generation are hampered by the absence of relevant information at a global level. Tegen and Fung [1995] attempted to estimate the effect of land use on dust mobilization on global scale and concluded that 20–50% of the global dust load was derived from “disturbed land.” However, Prospero *et al.* [2002] argued that TOMS AI suggests that most of the active sources are natural sources derived from topographic lows. Our previous study [Mahowald *et al.*, 2002] suggested that it may be difficult to determine how active disturbed source are in North Africa in comparison to satellite data because the sources are similar in geographic distributions, and the resulting optical depths overlap. In order to test if the disturbed dust source is active, here we carried out one sensitivity simulation of disturbed dust source. We use a very simple method to define the sources of mineral aerosols from cultivation: multiple a cultivation data set [Matthews, 1983] by predicted desert regions from the BIOME3 equilibrium vegetation model for the current climate [Haxeltine and Prentice, 1996], similar to Mahowald *et al.* [2002]. This source region then represents regions in which there is cultivation occurring in desert regions, thereby disturbing the soil. In addition, we add in the natural topographic low sources from the BASE case such that each contributes 50% to the atmospheric loading. Figure 1 shows a contrast between the source area used in the BASE simulation from *Ginoux et al.* [2001] and that used in the BASE+CULT sensitivity study, while Figures 1c and 1d contrast the mobilization resulting from these changes. Notice that the sources are very similar in geographic extent.

3. Results From 22-Year Simulation

[16] In this section we compare the results of the monthly mean simulations for a 22-year simulation to several available data sets. The overall budget of desert dust in the model is shown in Table 2, and shows that the model predicts a

Table 2. Annual Budget Averaged From 1979 to 2000 for Each Size Class

Size, μm	Emissions, Tgyr^{-1}	Dry Deposition, Tgyr^{-1}	Wet Deposition, Tgyr^{-1}	Atmospheric Loading, Tg	Lifetime, day
.1–1	165	4.9	161	4.7	10.3
1–2.5	496	63	433	12.5	9.2
2.5–5	496	338	156	4.5	3.3
5–10	496	418	48	1.5	1.1
.1–10	1654	823	798	23	5.1

global dust source of 1654 Tgyr^{-1} , with an average lifetime over the 4 bins included in the study of about 6 days, with the deposition split evenly between wet and dry deposition. Note that the 22-year simulation uses the BASE case.

3.1. Optical Depth

[17] In this study, we use optical properties for dust as described by Zender *et al.* [2003] and the visible indices of refraction ($n = n_r + in_i = 1.56 + 0.0038i$) for dust measured by Patterson [1981]. The extinction was calculated on a high-resolution size grid, then weighted by the appropriate sub-bin distribution and integrated to bin-mean values. The resulting extinction coefficients are consistent with available data [Zender *et al.*, 2003]; however there are large uncertainties in the optical properties of mineral aerosols, adding uncertainties to our comparisons with satellite data [e.g., Sokolik and Toon, 1996]. The optical depth is calculated from the dust mass load multiplied by mass extinction efficiencies. To evaluate modeled distributions of mineral aerosols, modeled mineral aerosol optical depths are compared against the AVHRR satellite data [Husar *et al.*, 1997], and MODIS satellite data (<http://modis-atmos.gsfc.nasa.gov>). Due to the large variability of the land surface reflectance in the visible range, only information over the ocean can be retrieved from the AVHRR instrument. Although the AVHRR instrument measures total aerosol optical depth (including other aerosols such as sulfates, black carbon, etc.), desert dust is a major aerosol component in several regions such as the tropical-subtropical North Atlantic, western Pacific, and Arabian Sea [e.g., Tegen *et al.*, 1997]. In other areas (such as southern Africa in June through November), the retrieved aerosol optical depths are likely to be dominated by other aerosols (such as biomass burning aerosols). Note that the AVHRR average represents optical depth under cloud free conditions, while model averages include all conditions. Figure 2 shows the comparison of seasonal mean optical depth between AVHRR and model simulations in which dust optical depth were calculated at same wavelength as the AVHRR observations (PATMOS2 at 630 nm), both averaged for 1989–1991. The comparison shows that the distribution of the model calculated optical depths coincide well with the observed values over the dust-dominated regions, such as the North Atlantic, Arabian Sea, and the western Pacific, although the model may be slightly overpredicting the maximum near North Africa. Most of the Southern Hemisphere does not appear to have strong aerosol optical depths in either the satellite retrievals or the model simulations. Seasonal variations in the maximum optical depth as a function of latitude over the North Atlantic and the Arabian Sea are realistically captured by the model.

[18] To utilize the newest generation of satellite data, we also compare the model calculated optical depth against MODIS observed optical depth in 2001 (when MODIS data becomes available for a full year). Figure 3 shows seasonal variation of optical depth from MODIS and model at 550 nm. The quality of retrieval aerosol optical depth is higher over ocean than over land, since more channels data were used over ocean than over land [Remer *et al.*, 2002; Chu *et al.*, 2002]. Again, the comparison of the distribution of the model calculated optical depths suggests that the model agrees with the observed values over the dust-dominated regions, such as North Atlantic, Arabian Sea, and East Asia, with a slight overprediction in the maximum again apparent close to North Africa. The model reproduced seasonal variation with a maximum in dust over North Atlantic and the Arabian Peninsula in summer. Similar to AVHRR, the large optical depths in the satellite over the North Pacific in spring and summer are likely to be caused by anthropogenic aerosols, while the southern Africa optical depths may be dominated by biomass burning aerosols, which are not included in our simulations.

[19] Next the model calculated optical depths are compared against the sun photometer Aerosol Robotic Network (AERONET) data [Holben *et al.*, 1998]. We chose several sites for comparison in regions where desert dust is considered to be the dominant aerosol type. The model calculated and AERONET observed optical depths both are at 670 nm. Two things should be noted for the comparisons. First, the optical depths were observed under cloud-free conditions, while the model calculations are total optical depths under both cloud-free and cloudy conditions. Second, the model optical depths are averaged from 1979–2000, and observations are from available data (mostly from 1996 to 2000). There is very little change in the comparison if model values from 1996–99 are used instead. Shading shows one standard deviation from the mean for the model simulations in Figure 4. Figure 4 shows that the comparisons are generally good, especially at Barbados (13°N , 59°W) and Mauna Loa (19°N , 155°W). But the model predictions of optical depth are higher than the observations at Sede Broker (26°N , 38°E), Bahrain (26°N , 50°E), and Banizoumbou (13°N , 2°E) from January to July; while the model simulation from Ginoux *et al.* [2001] tend to be in better agreement with the observations at these sites (although they use optical depths at 450 nm instead). Sede Broker and Bahrain are both located in Arabian Peninsula, which implies that the model may be over predicting the desert dust in Arabian Peninsula in spring and summer. Since we use the same source area as the Ginoux *et al.* [2001], it suggests that differences in meteorology, mobilization and/or deposition parameterizations are responsible for our differences over the Arabian Peninsula. Unfortunately our sensitivity study does not illuminate this issue (see discussion below).

3.2. Surface Observations

[20] There are in situ observations of monthly averaged desert dust concentrations available from 17 sites globally (courtesy of D. Savoie, J. Prospero, and R. Arimoto) [e.g., Prospero and Nees, 1986; Prospero, 1990; Prospero *et al.*, 1996; Arimoto *et al.*, 1990, 1997]. We compare against modeled values in Figure 5 (shading shows one standard

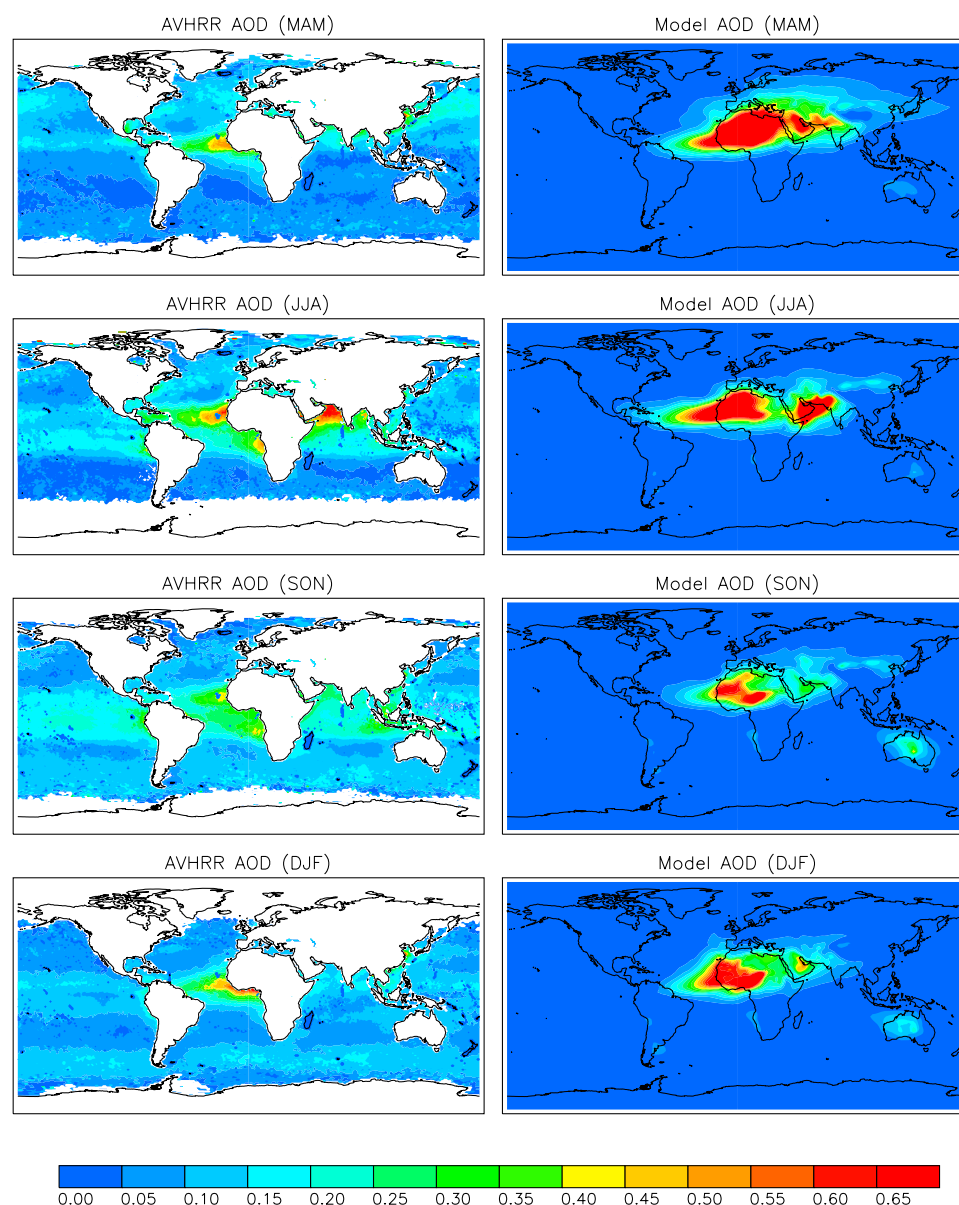


Figure 2. AVHRR observed and model calculated seasonal optical thickness at 630 nm averaged from January 1989 to December 1991 [Husar *et al.*, 1997].

deviation for model production). Dust concentrations were compared at same altitude between model and observation. In Figure 5 the observations and the model are for the same period only at Barbados (1979–2000), Capo Verde (1996–1997), Cheju (1991–1994), Hedo (1991–1994), and Mace Head (1991–1994). The comparisons at other sites used the simulation data (averaged from 1979–2000) and available observed data, at Bermuda (89–97), Cape Grim (88–91), Enewtak (81–86), Funafuti (83–86), Oahu (81–95), Izana (89–97), Miami (79–99), Midway (81–99), and Norfolk (83–86). The reason we use slightly different periods for the different sites is that we would prefer to compare to the model predicted concentrations averaged from 1979–2000 (our climatology), but for Capo Verde, Cheju, Hedo and Mace Head, the 22-year average and the average over the years when the observations take place are actually different enough to degrade our comparisons.

[21] It can be seen that the model captures the seasonal cycle of dust concentrations at most sites in the Northern Hemisphere, suggesting that the source mobilization and transport calculated by model are generally reasonable in the Northern Hemisphere. The model slightly over predicts concentrations close to the North African sources at Izana and Capo Verde. At Cheju and Hedo, the model under-predicts concentrations during the early part of the year, similar to *Ginoux et al.* [2001] and *Tegen et al.* [2002]. Our model appears to accurately simulate the mineral aerosols transported from East Asia to the Pacific Islands (Midway and Oahu) better than *Ginoux et al.* [2001], and does not require vegetation phenology, as suggested by *Tegen et al.* [2002] in order to so. Transport from East Asia to the Pacific is very strong during the spring months and this controls the strong seasonal cycle seen at Hawaii in many constituents (MLOPEX experiments). In the Pacific, closer

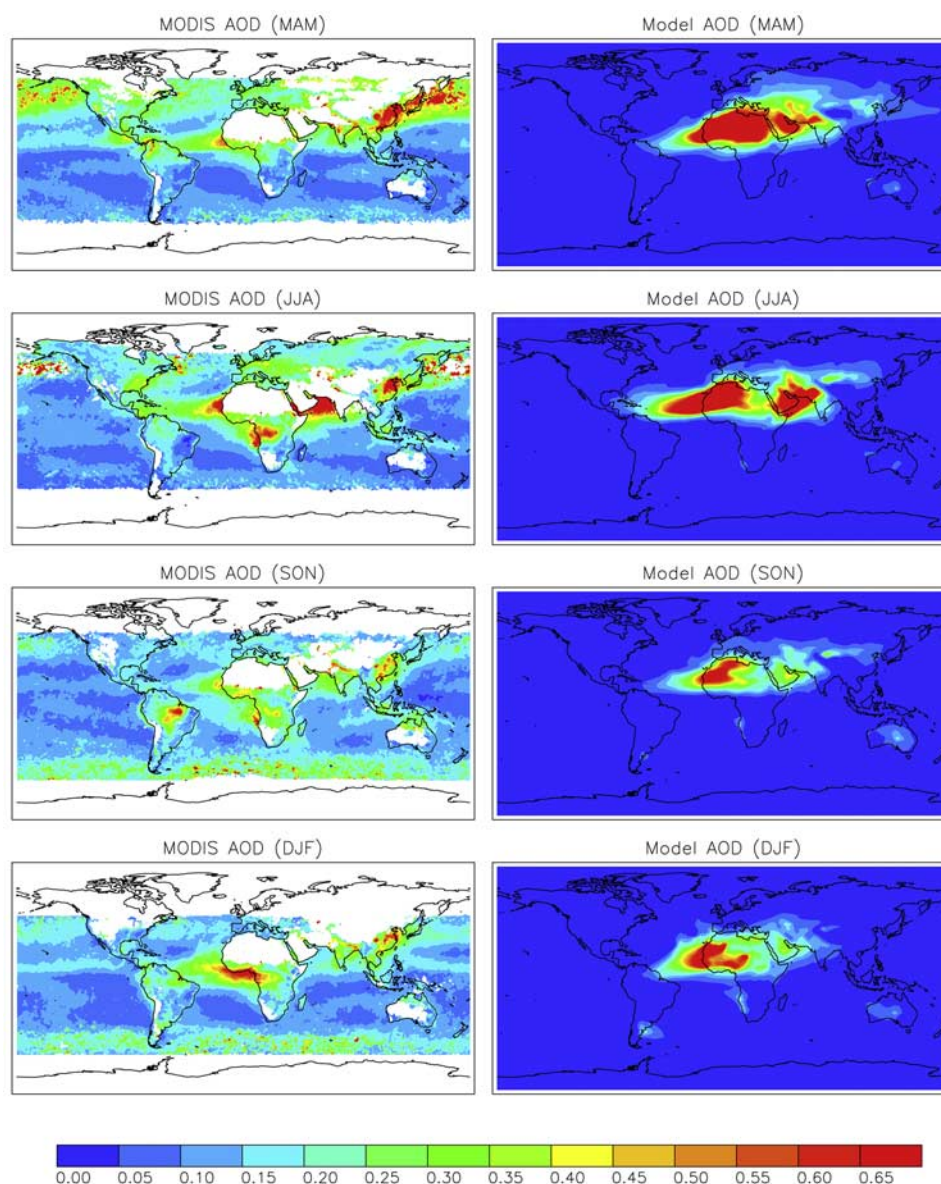


Figure 3. Optical depth comparison between MODIS measured and model calculated at 550 nm in 2001 (<http://modis-atmos.gsfc.nasa.gov>).

to the dateline, the model appears to simulate the concentrations at Enewtak well, but on the other side of the equator, this model simulation underestimates the desert dust carried to Funafuti (8.5°S, 179.2°W), which may be due to too strong of removal, or inadequate transport; this poor comparison is similar to *Tegen et al.* [2002], while the modeling study of *Ginoux et al.* [2001] compares more favorably at this station.

[22] Unfortunately, the model simulation in the Southern Hemisphere is not as realistic as in the Northern Hemisphere: the model over-predicts surface dust concentration at Cape Grim and New Caledonia, and simulated Norfolk fairly. The *Ginoux et al.* [2001] or *Tegen et al.* [2002] simulations compare much better to the observations at the Southern Hemisphere stations. Since the observations at Cape Grim may include some local sources not included in

the model, these observations should be considered an upper bound, and thus our over prediction at this site is likely to be even larger than seen in Figure 5. Differences between the *Ginoux et al.* [2001] simulation and this simulation are not due to differences in the source areas (since they are the same), but rather to differences in meteorology (they use the Goddard Earth Observing System Data Assimilation System (GEOS DAS) analysis), the mobilization scheme, and/or wet deposition processes. *Tegen et al.* [2002] have slightly different source areas and use the European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis. Thus it appears that concentrations in the Southern Hemisphere are not only sensitive to limiting the area of the sources to just dry topographic lows as proposed by *Prospero et al.* [2002] and argued by *Ginoux et al.* [2001], but that Southern

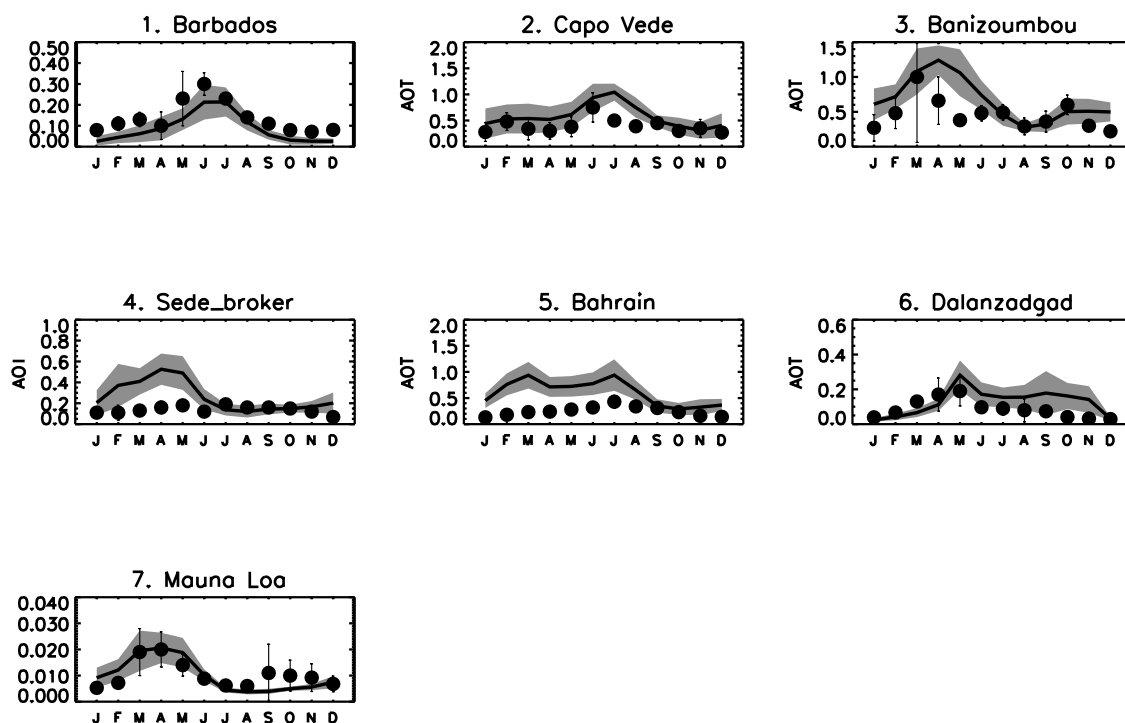


Figure 4. Comparison of optical thickness (at 670 nm) between AERONET (dots) and model simulation (solid line). Shading show one standard deviation for model production. The simulated values are averaged from 1979 to 2000, but the observed values are averaged from available data (mostly from 1996 to 2000 [Holben *et al.*, 1998]).

Hemisphere distributions are very sensitive to the meteorology and exact parameterization of the mobilization and sinks (see Section 4).

3.3. Deposition

[23] In order to compare the more accurate in situ deposition data to our model results, we show a scatterplot comparison of annual deposition similar to Ginoux *et al.* [2001] (Figure 6). The site locations and periods of the measurements are shown by Ginoux *et al.* [2001, Table 6]. The range of observed values varies from $450 \text{ gm}^{-2}\text{yr}^{-1}$ over the Taklimakan to $0.08 \text{ gm}^{-2}\text{yr}^{-1}$ in the equatorial Pacific. Overall the model compares well with the observations, except at the Taklimakan (site 15, 40°N , 85°E), Nauru (site 9, 53°S , 167°E), New Zealand (site 14, 35°S , 173°E), and French Alps (site 2, 46°N , 7°E) sites. The model over prediction of deposition in Southern Hemisphere sites (Nauru and New Zealand) is consistent with the over prediction of atmospheric desert dust concentrations at several Southern Hemisphere stations shown in Figure 5. Since the Taklimakan site is located close to an

active dust source, the large observed deposition could have substantial contribution from very large particles (larger than $10 \mu\text{m}$), which are not included in the model simulation. In addition, the discrepancies between the model and observations could be due to the coarse grid of the global model being unable to resolve important processes, especially in mountainous regions. Because of the short lifetime of larger particles to dry deposition, the deposition rates close to the source areas will be very sensitive to the exact location of the source areas. Overall, the model compares well with available deposition data, similar to Ginoux *et al.* [2001].

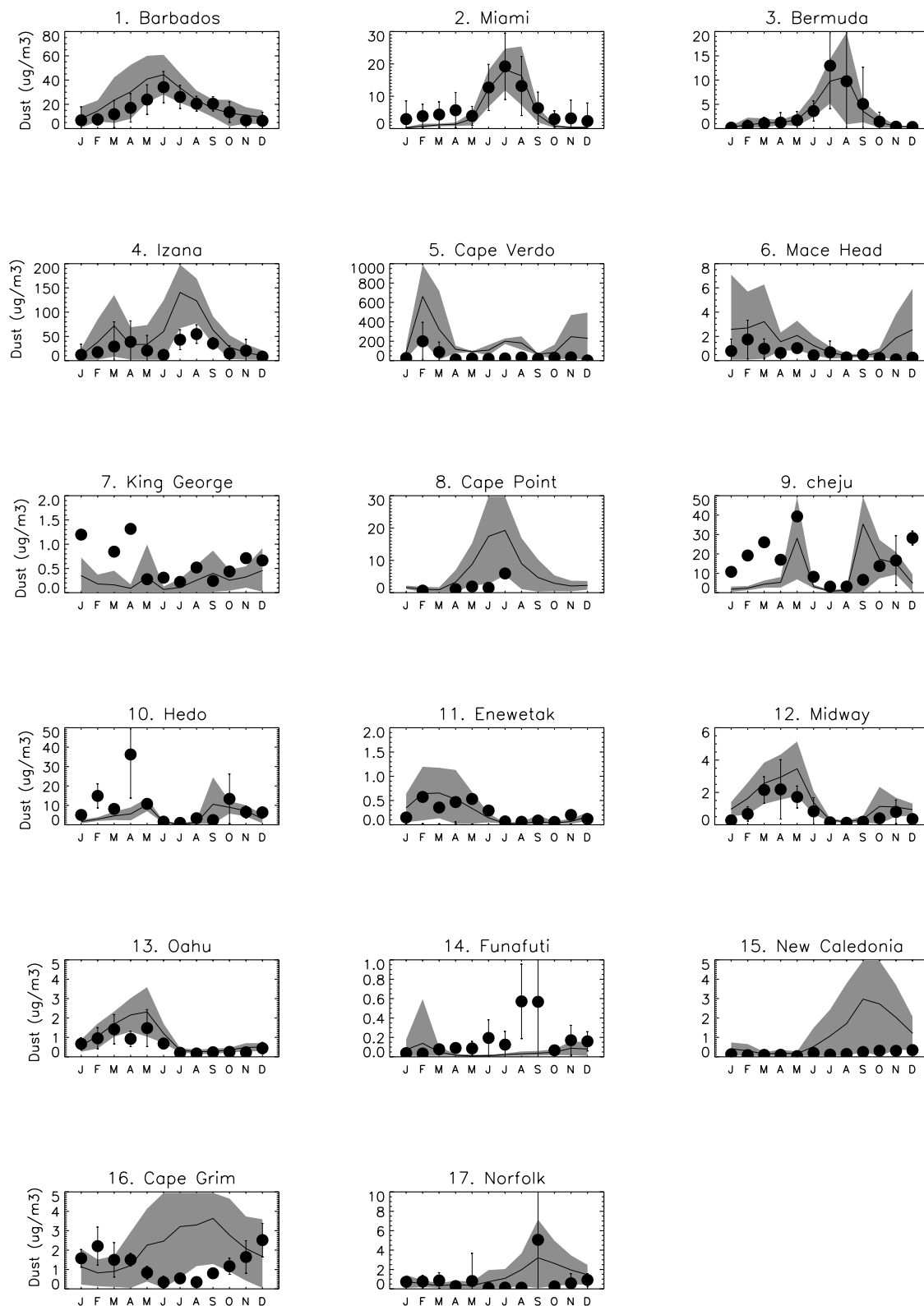
3.4. Mobilization

[24] In the parameterization described in above, the mineral aerosol mobilization strongly depends on surface winds, atmospheric stability, soil moisture, and vegetation. We briefly explore which factor exerts the strongest control in our control model simulation in the North African, Arabian Peninsula, East Asian, Australian and Sahel source areas considering area averages in our 22-year base simu-

Figure 5. (opposite) Comparison of monthly mean dust concentrations, simulated (line) and observed (circles) at 17 sites, in units of μgm^{-3} . Simulated values are averaged from 1979 to 2000 (with exceptions listed in the text), but observed values are averaged from available data. Observations are courtesy of D. Savoie, J. Prospero, and R. Arimoto. Shading shows one standard deviation for model production. Site location: Barbados (13.17°N , 59.43°W), Miami (27.75°N , 80.25°W), Bermuda (32.27°N , 64.87°W), Izana (28.30°N , 16.50°W), Cape Verde (16.70°N , 22.90°W), Mace Head (53.32°N , 9.85°W), Cheju (33.52°N , 126.48°E), Hedo (26.92°N , 128.25°E), Cape Grim (40.68°S , 144.68°E), Enewtak (11.33°N , 162.30°E), Funafuti (8.50°S , 179.20°E), Norfolk (29.08°S , 167.98°E), Midway (28.22°N , 177.35°W), Oahu (21.33°N , 157.70°W), King George (62.18°S , 58.30°W), New Caledonia (22.15°S , 167.00°E), Cape Point (34.35°S , 18.48°E).

lation. Of course, spatial and temporal heterogeneities are important in desert dust emissions; however an averaged analysis of the entire region provides a simplified picture of what meteorological factors control the sources in our rather complicated dust module. Figure 7 shows the normalized monthly averaged values of the dust mobilization, friction

velocity, wind velocity at 10 m, precipitation and soil moisture in North Africa (10°N–35°N, 17°W–30°E), Arabian Peninsula (12°N–26°N, 39°E–47°E), East Asia (39°N–43°N, 84°E–111°E), Australia (22°S–31°S, 122°E–141°E), and Sahel (5°N–20°N, 0°–40°E). We normalize the values to plot on the same vertical axis, by



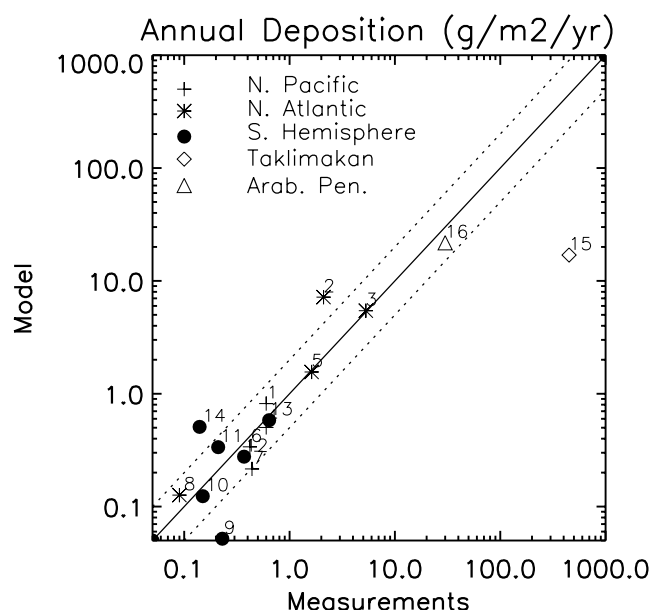


Figure 6. Comparison of annual total deposition flux simulated and observed at 16 sites, in units of $gm^{-2}yr^{-1}$ (compiled deposition data from *Ginoux et al.* [2001]). Site location: 1. Shemya (52.92°N, 174.06°E), 2. Fench Alps (45.5°N, 6.5°E), 3. Spain (41.8°N, 2.3°E), 4. Midway (28.2°N, 177.35°W), 5. Miami (25.75°N, 80.25°W), 6. Oahu (21.3°N, 157.6°W), 7. Enewetak (11.3°N, 162.3°E), 8. Fanning (3.9°N, 159.3°W), 9. Nauru (0.53°S, 166.95°E), 10. Samoa (14.25°S, 170.6°W), 11. Rarotonga (21.25°S, 159.75°W), 12. New Caledonia (22.15°S, 167.0°E), 13. Norfolk Island (29.08°S, 167.98°E), 14. New Zealand (34.5°S, 172.75°E), 15. Taklimakan (40.0°N, 85.0°E), 16. Tel Aviv (32.0°N, 34.5°E).

subtracting the annual mean and dividing by the standard deviation. The correlations between different variables are shown in Table 3. Clearly, winds at 10m and wind friction velocity are highly correlated (correlation coefficients are over 0.95 at the source areas), and dust mobilization and wind velocity at 10m or wind friction velocity are correlated highly (between 0.96 and 0.99), suggesting that the difference between using the friction velocity (here or *Tegen et al.* [2002]) or the winds at 10 m [e.g., *Ginoux et al.*, 2001] may not be important for the mobilization. Notice that if we used maximum sustained winds we might obtain higher correlations (more similar to *Tegen and Miller* [1998]) because of the nonlinear relationship between mobilization and winds, but we chose here to compare against the monthly mean winds, since these are commonly archived from numerical weather prediction and general circulation models. As expected, the amount of precipitation and the soil moisture are moderately to highly correlated. There is negative correlation between dust mobilization and soil moisture except at the Arabian source area, probably due to the small positive correlation between soil moisture and wind friction velocity ($r = 0.36$). The soil moisture is more important in Sahel than in North Africa. The averaged analysis suggests that in our model, the mobilization is very strongly controlled by the winds, with small modifications due to the soil moisture (except in East Asia and Sahel where winds

and soil moisture are equally important), emphasizing that wind data quality in the source regions may control the model simulation quality.

3.5. Source Apportionment Study

[25] In order to estimate the modeled contributions of different dust source regions to the overall dust loading, concentrating on the Northern Hemisphere, we define an East African source (10°N–35°N, 17°E–30°E), Central African source (10°N–35°N, 0–15°E), West African source (10°N–35°N, 0–17°W), South Arabian Peninsula source (10°N–26°N, 43°E–58°E), East Asian source area (31°N–47°N, 75°E–114°E), and Australia source (26°S–33°S, 118°E–142°E), and simulate the distributions from each source for the year 1998 by using BASE compared against total dust loading for the same year. Figure 8a shows each source area included in the model sensitivity analysis. Figures 8b–8h show the distribution of the ratio of the annual average dust loading from each source divided by the annual average total dust loading as predicted by the model. Tables 4 and 5 show the budget, and the column loading over different ocean regions for each source, respectively (notice that the percentage dust loading in each column will sum to $\sim 100\%$). Clearly the North Africa dust sources (especially Central North Africa) not only control North Atlantic dust distribution, but also have the largest contribution to the Northern Hemisphere and global dust loadings in this model, and the Australian source controls South Hemisphere dust loading (see Figure 8 and Tables 4 and 5). In Table 4 we see that the African sources account for 67% of the total global dust mobilization and 73% of the total dust loading in the model. Notice that the wet deposition lifetimes vary by almost a factor of two between different regions (7–13 days), with the longest lifetimes from the North African sources in Table 4. The global estimate of wet deposition lifetimes will be strongly controlled by the North African lifetime because of the large emissions in that region. Table 5 and Figure 8 show the calculated percentages of dust loading in different regions from each source relative to all sources according to model calculations. The model predicts that the North Atlantic dust loading is controlled by the West, Central, and East Africa sources, accounting for 27%, 47% and 11% of the loading, respectively. North Pacific dust is dominated by the East Asian source (15%), East African source (14%), Central African source (30%), and West African source (15%), with some impact from the miscellaneous sources (mostly in Central Asia). Thus, the Africa dust sources have a large contribution not only to the North Atlantic but also to the North Pacific. This model simulation may overestimate the strength of the North African source relative to the East Asian source, since ice core provenance analysis in Greenland for the current climate suggests that the East Asian source dominates deposition there, especially during the springtime [*Bory et al.*, 2003]. The northern Indian Ocean has a large contribution from the Central Asian sources (which we do not simulate separately), and the South Arabian Peninsula source (26%) and the African sources (10% from East African, 13% from Central African and 6% from West Africa). South Hemisphere dust is dominated by the Australia source. South Pacific, South Atlantic, and South Indian Ocean dust were contributed by Australia

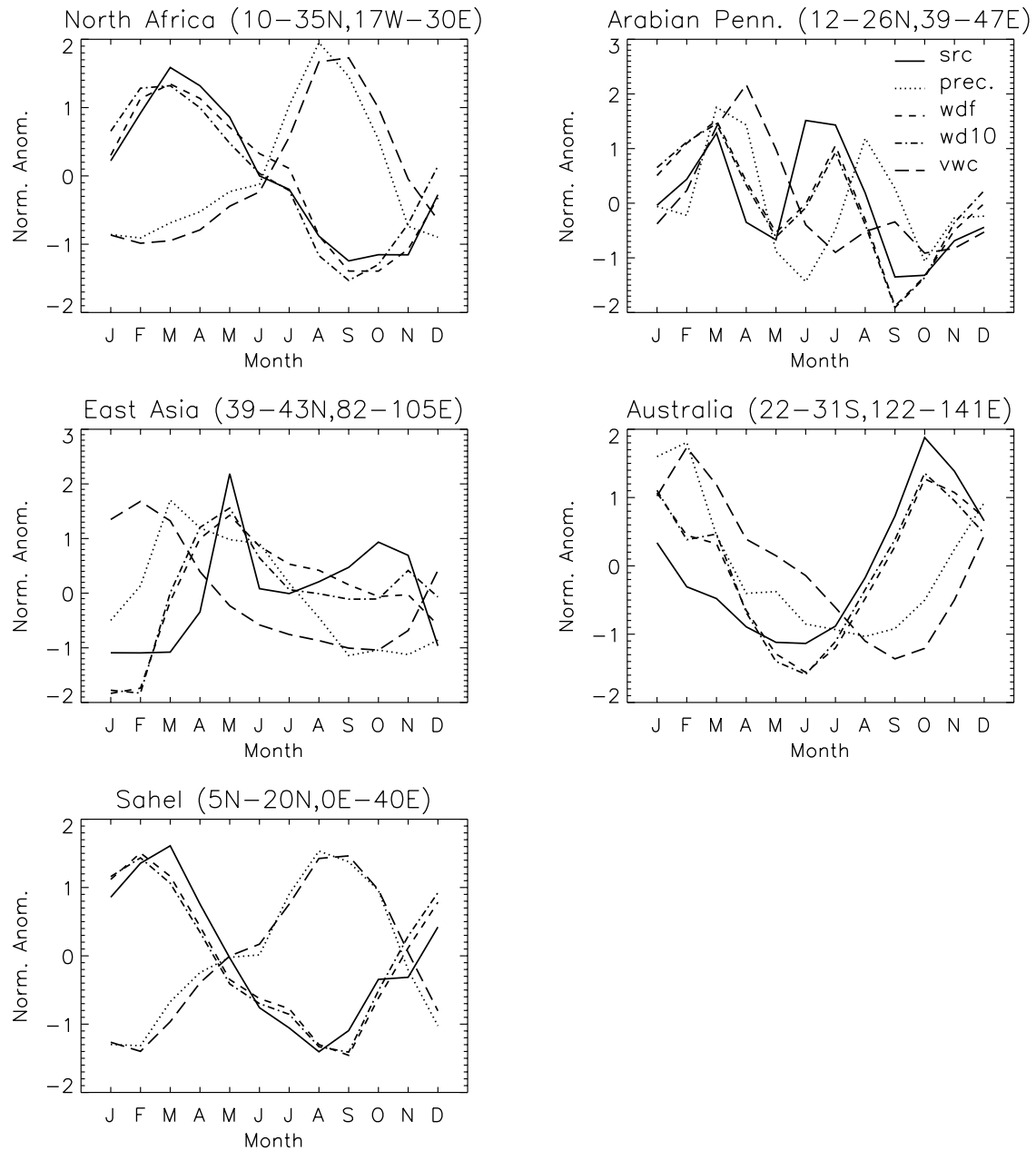


Figure 7. Normalized monthly average source (src), precipitation (prec), friction wind velocity (wdf), wind velocity at 10 m (wd10), and soil moisture (vwc) in North Africa (a), Arabian Peninsula (b), East Asia (c), Australia (d), and Sahel (e), averaged from 1979 to 2000. Normalized values are calculated as (monthly mean_{annual mean})/monthly standard deviation.

Table 3. Correlation Between Different Meteorological Parameters in Selected Source Areas^a

Source Area	Soil Moisture Versus Precipitation	Soil Moisture Versus Mobilization	Wind (10 m) Versus Friction Wind Velocity	Friction Wind Velocity Versus Mobilization	Wind (10 m) Versus Mobilization
N. Africa	0.94	−0.79	0.96	0.98	0.94
Arabian Penn.	0.57	0.09	0.99	0.79	0.75
E. Asia	0.40	−0.72	0.95	0.69	0.65
Australia	0.82	−0.37	0.99	0.87	0.86
Sahel	0.97	−0.91	0.99	0.95	0.93

^aAll correlations are significant at 95% confidence level, except the correlation between soil moisture and mobilization in Arabian Peninsula. The differences of two correlations between soil moisture versus source and friction wind speed versus source are significant in Arabian Peninsula and Australia source regions.

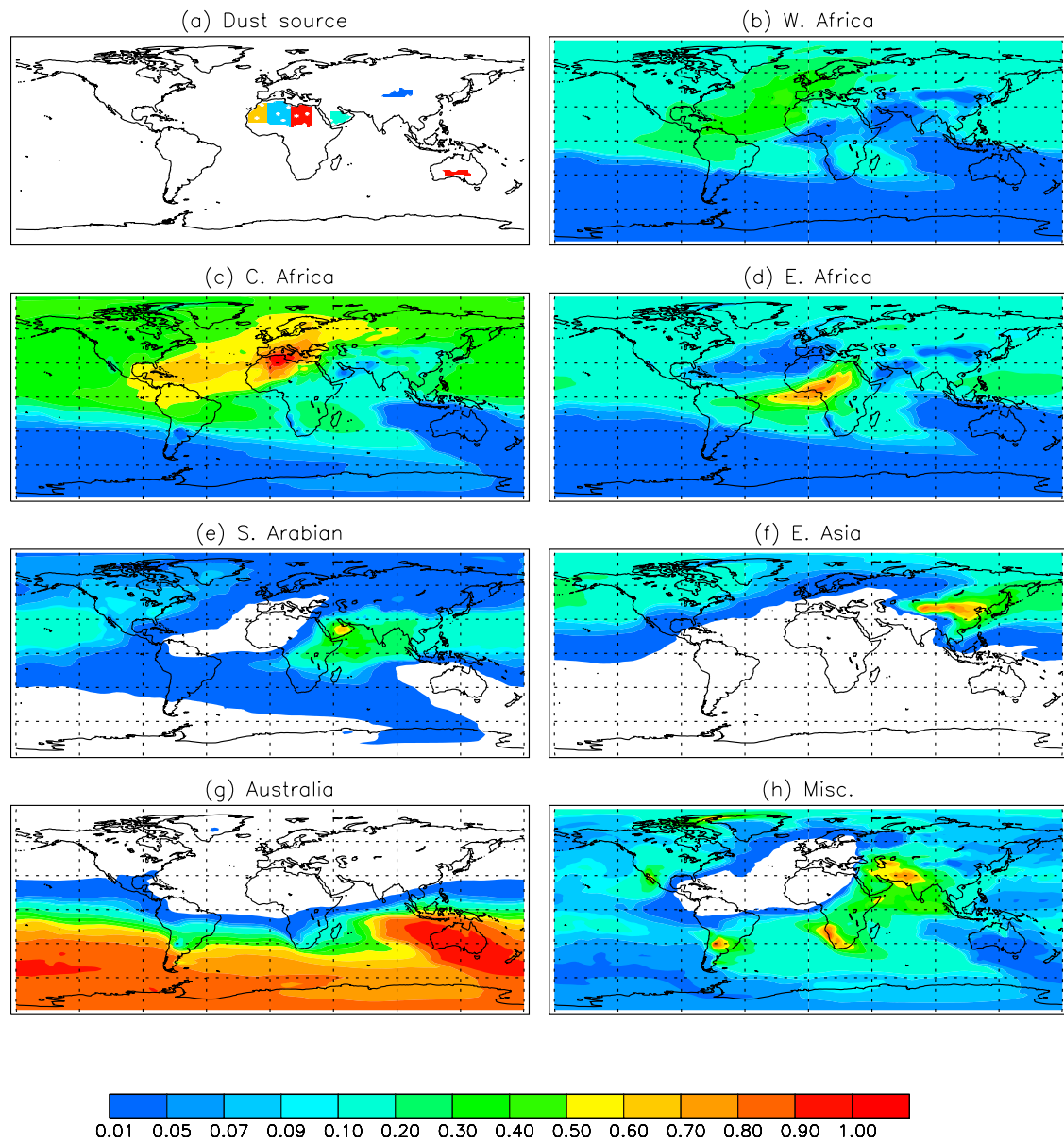


Figure 8. Source apportionment study, where sources are shown in different colors in (a) west North Africa in yellow, central North Africa in blue, east North Africa in red, South Arabian Peninsula in green, and East Asian source in blue. Ratio of annual average dust loading for each source area divided by total annual average dust loading from the model simulation in 1998 for west North Africa (b), central North Africa (c), East North Africa (d), southern Arabian Peninsula (e), East Asia (f), Australia (g), and miscellaneous sources (all others) shown in (h).

Table 4. Annual Budget of Each Source Area for Sensitivity in 1998

Source Area	Emission, T_{gyr}^{-1}	Dry Deposition, T_{gyr}^{-1}	Wet Deposition, T_{gyr}^{-1}	Atmos. Loading, T_g	Dry Lifetime, day	Wet Lifetime, day
W. Africa	257	136	115	3.82	10.2	12.1
C. Africa	544	293	238	8.71	10.8	13.4
E. Africa	313	158	148	4.27	9.7	10.6
Arabian Peninsula	119	57	59	1.56	9.9	9.6
E. Asia	54	24	29	0.64	9.5	7.9
Australia	132	70	59	1.69	8.8	10.4
Misc. source	235	86	150	2.46	10.4	6.0

Table 5. Dust Loading Contributions From Each Source in Each Region^a

Source	North Atlantic	South Atlantic	North Pacific	South Pacific	North Indian Ocean	South Indian Ocean
W. Africa	27	8	15	6	6	6.5
C. Africa	47	16	30	11	13	12
E. Africa	11	17	14	5	10	9
Arabian Peninsula	4	3	9	2	26	6.5
E. Asia	4	0.14	15	0.3	1	0.3
Australia	0.6	40	3	69	2	48
Misc. source	6.4	16	14	7	42	17.7

^aUnits are in percent. Source location: North Atlantic (1°N–60°N, 21°W–103°W); South Atlantic (0°–58°S, 0°–60°W); North Pacific (1°–60°N, 129°E–128°W); South Pacific (0°–58°S, 105°E–84°W); North Indian Ocean (1°–18°N, 54°E–94°E); South Indian Ocean (0°–58°S, 41°E–99°E).

source 69%, 40%, and 48%, respectively, with miscellaneous sources making up the remainder of the dust. It is likely that other modeling studies will have different relative source importance and it would be interesting to contrast these in the future.

4. Sensitivity Studies

[26] In order to understand the differences between the model simulations presented here and by *Ginoux et al.* [2001], we conduct several sensitivity studies using different source parameterization and meteorology data sets. These studies are conducted for January through November 1995, and the model scenarios are described in more detail in Section 2.2.

4.1. Comparison Between NCEP/NCAR and DAO/NASA Reanalysis Data

[27] Since the sensitivity simulations compared here are based on two different meteorological data sets, we first briefly compare several main parameters (surface wind speeds, precipitation and planetary boundary layer high) between NCEP and DAO data sets. Table 6 shows the correlation coefficients and linear fit parameters between January monthly averaged DAO and NCEP on a spatial basis. Note that results are similar in other months. All comparisons show the correlation coefficients of monthly averaged parameters of two data sets are higher than 0.79 (see Table 6). Note that for low values of surface winds, precipitation, and planetary boundary layer height, NCEP tends to have higher values than DAO (notice the intersect is nonzero). Overall, the data sets are consistent, however, as expected.

[28] Figure 9 shows spatial distribution of precipitation between archived and MATCH calculated from NCEP and DAO. It can be seen that MATCH.DAO and MATCH.NCEP both get monthly precipitation highest in the same regions (the results are similar in other month, not show here). The regions with larger rain are in Southeast Asia, North and South Pacific, Indian Ocean, South Africa, South America, and North Atlantic.

[29] Next we compare the precipitation predicted in MATCH, from the reanalyses archives, and from observations [*Xie and Arkin*, 1996]. Spatial correlations between monthly averaged MATCH-NCEP and NCEP-archived or

between MATCH-DAO and DAO-archived precipitation are over 0.9, suggesting that MATCH is capable of reproducing the reanalyses precipitation and thus the convective mixing in DAO data sets (as well as ECMWF and NCEP [*Mahowald et al.*, 1997]). However, there are substantial errors in the reanalysis precipitation, since correlations between the NCEP or DAO precipitation and observed precipitation [*Xie and Arkin*, 1996] are between 0.25 and 0.30, consistent with previous studies of the hydrological cycle in reanalyses [e.g., *Trenberth and Guillemot*, 1998; *Santer et al.*, 2000].

[30] Next we use radon222 as a diagnostic for differences in vertical transport between the NCEP and DAO data sets [e.g., *Mahowald et al.*, 1997]. Radon222 decays with a 5.5 day e-folding time scale, and has a relatively constant continental source, which we assume is constant everywhere [*Jacob et al.*, 1997]. Figure 10 shows the radon vertical distribution in November over 4 dust source regions. The radon vertical distribution suggests that the vertical transport of radon is roughly similar using the NCEP and DAO data sets, especially in East Asia. In North Africa and the Arabian Peninsula, there is more radon in the DAO simulation than in the NCEP simulations between 950 and 600 mb, but slightly more in NCEP simulations at 200–300 mb level. Over Australia, the reverse appears to be true, where the NCEP simulation has more mixed more radon between 950–600 mb, but the DAO simulation has more radon above 600 mb. These differences are relatively small compared to the intercomparisons [e.g., cf. *Jacob et al.*, 1997].

4.2. Results From Sensitivity Studies

[31] Analysis of the wet and dry deposition lifetimes for different regions suggest that the lifetimes are similar for the different cases (within 10%) and even regionally (Table 7). The residence time of dust in over the Eastern Pacific region is almost 2 times the global average for all sensitivity analyses simulations (see Table 7). The dust impact on radiation or chemistry can thus be particularly important in this region. The residence time of dust is shorter than the global average in south Atlantic due to faster wet removal. Table 8 shows the budget of each sensitivity simulation. We normalized the global total source for all sensitivity analyses to 1682 Tg for 11 months. The lifetime of dust varies from 5.0–5.7 days. Dry deposition tends to be lower in simulations using the DAO meteorology (GDD and Base-DD) runs, perhaps due to the small differences in precipitation or winds seen in Section 4.1. The lifetime of dry and wet deposition are 10.2 and 10.8 days, respectively, in our BASE sensitivity simulation, which are much similar to our 22-year climatological simulation (dry lifetime

Table 6. Spatial Correlation Between DAO and NCEP Data Sets for January 1995^a

Month	A	B	R
Surface winds	0.237 ± 0.0219	0.868 ± 0.00398	0.85
Precipitation	0.408 ± 0.0130	1.008 ± 0.0042	0.87
PBLH	142.515 ± 2.409	0.77 ± 0.00448	0.79

^aWind units are in m s⁻¹, precipitation is in m month⁻¹, and PBLH is in meters. $Y = a + bX$, Y is NCEP monthly averaged data, and X is DAO monthly averaged data.

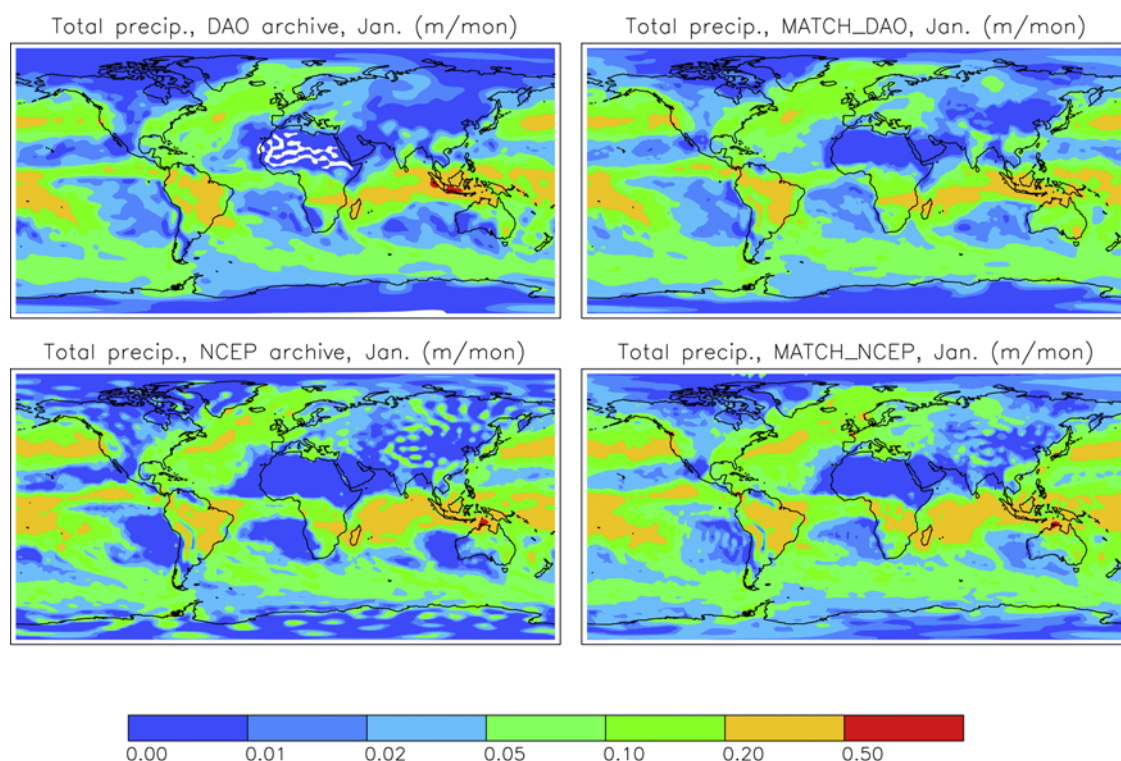


Figure 9. Precipitation spatial distribution between Archived and MATCH calculated from NCEP and DAO data, in units of $mmonth^{-1}$.

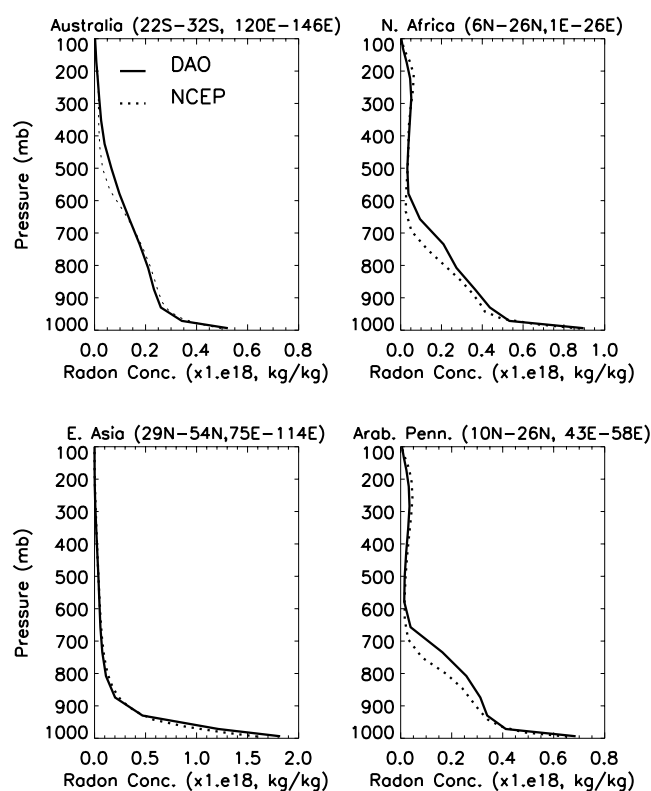


Figure 10. Radon vertical distribution over 4 dust source regions for November 1995.

10.2 days, and wet lifetime 10.5 days). The dry lifetime in our BASE simulation is longer than in *Ginoux et al.* [2001] (8.1 days), while the wet lifetime in our simulation is much lower than *Ginoux's* simulation (56 days) [*Ginoux et al.*, 2001], which is longer than most other predictions of aerosol lifetimes [e.g., *Balkanski et al.*, 1993; *Kriedenweis et al.*, 1999; *Guelle et al.*, 2000]. The large discrepancy between the lifetimes predicted in this model, and those used in the *Ginoux et al.* [2001] study, both of which match available observations, suggest that much work is necessary to improve our understanding of mineral aerosol wet removal processes.

4.2.1. Optical Depth

[32] Figure 11 shows distribution of annual mean dust optical depth for the different simulations. We note that the BASE case has larger optical depths over Australia and smaller optical depths over East Asia, similar to results (in Section 3.1) comparing our 22-year climatology and the

Table 7. Dust Lifetime for Different Sensitivity Analyses in Different Regions^a

	BASE	GDD	BASE-DD	GND	BASE+Cult
W. Pacific	9.1	9.4	9.5	9.6	9.4
E. Pacific	12.1	13.4	13.6	13.7	13.4
S. Pacific	4.5	4.9	4.9	4.9	4.9
N. Atlantic	6.5	6.4	6.5	6.5	6.4
S. Atlantic	3.2	3.7	3.7	3.7	3.6
Indian Ocean	5.7	6.9	6.7	6.7	6.9

^aRegion location: west Pacific (5°N–62°N, 144°E–174°W); east Pacific (5°N–62°N, 173°W–126°W); South Pacific (0°–62°S, 144°E–174°W); North Atlantic (6°N–65°N, 60°W–0°); South Atlantic (0°–62°S, 60°W–0°); Indian Ocean (20°N–37°N, 45°E–109°E).

Table 8. Dust Budget for Different Sensitivity Calculations in 1995^a

	Source, Tg	Dry Dep., Tg	Wet Dep., Tg	Burden, Tg	Lifetime, day
BASE	1682	847	805	26	5.2
GDD	1682	751	903	29	5.7
BASE-DD	1682	703	959	29	5.7
GDN	1682	801	843	29	5.7
BASE+CULT	1682	817	843	25	5.0

^aSource, deposition, burden, and lifetime are for 11 months from January to November 1995.

Ginoux et al. [2001] results. The differences in East Asia appear to be due largely to differences in meteorology, since when we use DAO winds in the BASE model, we obtain similar optical depths to those seen in the GDD case (contrast Figures 11a, 11b, and 11c). If we calculate the total dust mobilization in East Asia for the BASE, GDD and BASE-DD cases, we obtain 38.6, 180 and 173 Tg in 11 months, respectively, suggesting that the main differences in optical depth seen between our model simulations and *Ginoux et al.* [2001] are due to differences in surface meteorology near the source regions. More detailed calculations suggest this difference in mobilization between BASE and BASE-DD is dominated by differences in surface winds. We can see that transport differences are not responsible for the majority of the differences seen in the optical depth when we compare GDD versus GDN (Figures 11b and 11d).

[33] Optical depths over Australia are highest in BASE, next highest in BASE-DD and lowest in either GDD or GDN. The differences appear to be due to mobilization (since GDD and GDN appear similar over Australia). The mobilizations over Australia source area are 98, 46 and 19 Tg in 11 months for the BASE, BASE-DD and GDD cases, respectively, suggesting that both surface meteorology and source parameterization are important in Australia. More analyses show that the difference between BASE and BASE-DD appears to be mostly surface wind driven, with some contribution from difference of soil moisture.

[34] Optical depths over the Arabian Peninsula were too large in the BASE case climatology and smaller in the *Ginoux et al.* [2001] climatology. Unfortunately, in our simulations here, we do not reproduce this signal over the Arabian Peninsula in BASE and GDD, suggesting that processes we are not testing for (e.g., transport or deposition differences) may be responsible for the differences in this region.

[35] The addition of a cultivation source (BASE+CULT) increases the optical depths over India (Figure 11a versus Figure 11e) and reduces optical depths over Australia, in addition to changing the geometry of the sources close to the source areas in North Africa and Asia. Overall, the differences between including a 50% cultivation source and not including it are similar in magnitude to differences due to meteorological data sets (contrast Figure 11a versus Figure 11e and Figure 11a versus Figure 11b).

4.2.2. Surface Observations

[36] Next we explore the sensitivity of surface concentrations at the observing stations to meteorology and source parameterizations. Figure 12 shows the comparison of monthly averaged surface dust concentrations at 17 sites

in 1995. It is worth mentioning that observations at some sites are during the same year as the simulations (1995), but most observations are not the same year (shown in Figure 12). First we compare BASE and GDD cases. The monthly averaged surface dust concentrations are reasonable at the Northern Hemisphere sites for the BASE and GDD cases. Modeled surface dust of BASE run are more similar to observations at Barbados, Miami, Bermuda, and Oahu than the results of GDD run, but surface dust calculated by GDD concentrations are more similar to observations than BASE

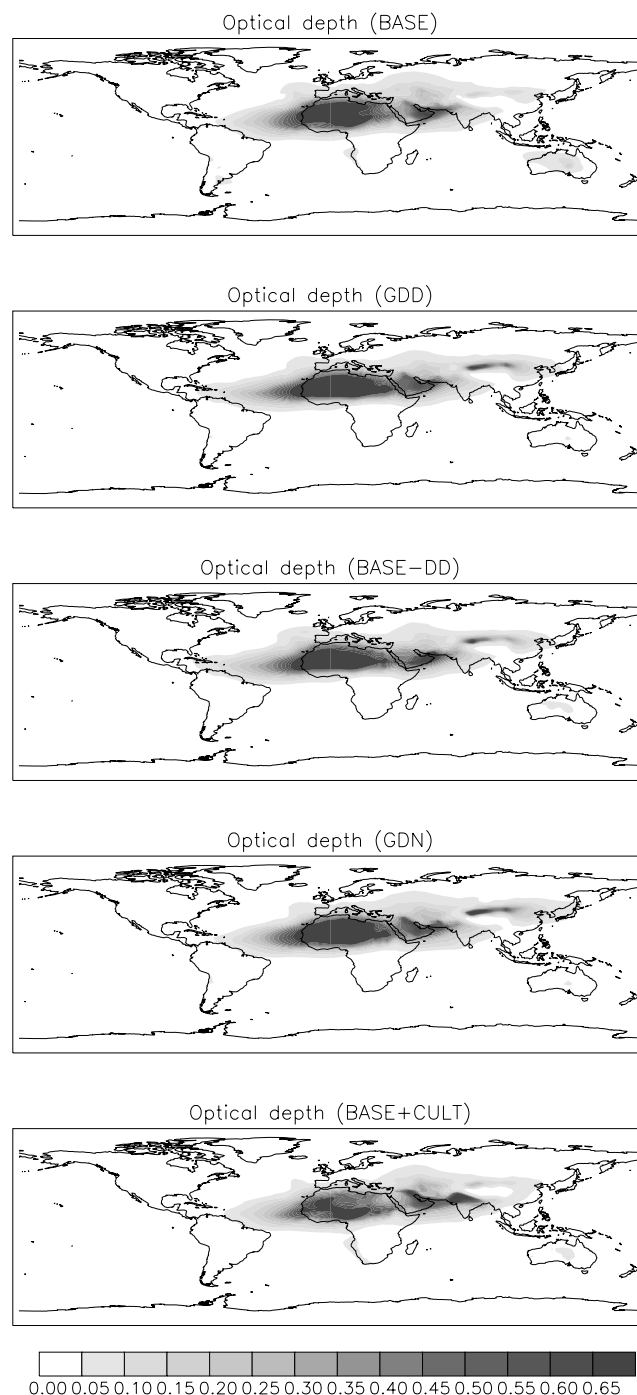


Figure 11. Annual mean dust optical depth (630 nm) distribution for different sensitivity runs.

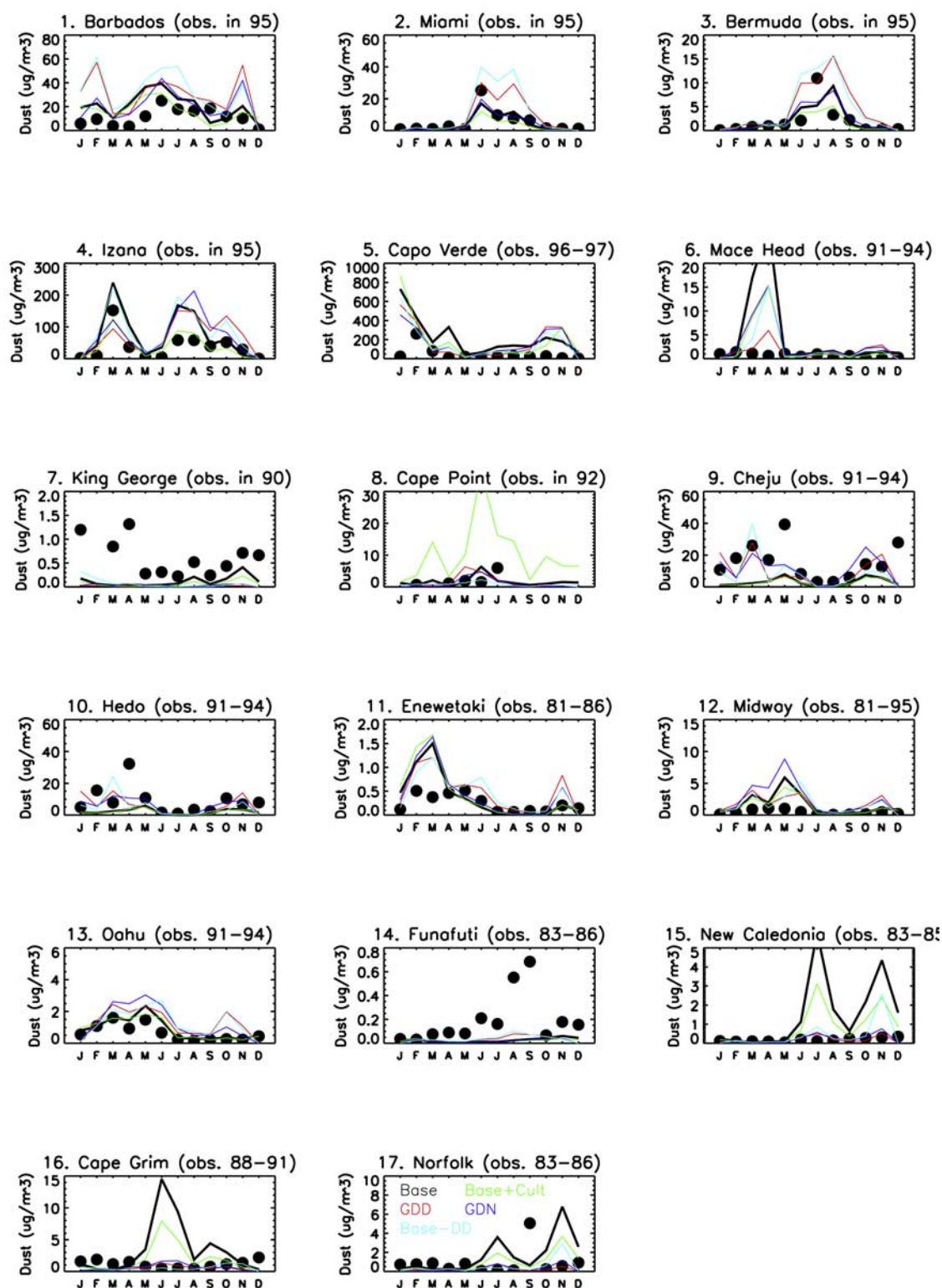


Figure 12. Comparison of monthly mean dust concentration, sensitivity simulated (lines), and observed (circles) at 17 sites, in units of $\mu\text{g m}^{-3}$ in 1995. Observations are courtesy of D. Savoie, J. Prospero, and R. Arimoto. The site locations are the same as in Figure 5.

at New Caledonia and Cape Grim in South Hemisphere. The biggest difference between the two sensitivity runs are in the South Hemisphere, as seen in the climatologies [Ginoux *et al.*, 2001] and our simulations (Figure 5), suggesting that our

sensitivity study should provide insight into the differences between Section 3.0 and Ginoux *et al.* [2001].

[37] In the climatologies, the East Asian source was stronger in Ginoux *et al.* [2001] than in DEAD/NCEP/

MATCH, and the comparisons of the mobilization suggested this was due to changes in meteorology in the source areas. However, it can be seen by comparing model results at Oahu that transport is important at this site. GDD and BASE-DD are higher than the BASE case (consistent with meteorology in the source region being important), but GDN is even higher than GDD, suggesting that NCEP winds transport the dust more efficiently to Oahu. This is also consistent with the large difference in optical depth (and mobilization) over the source regions between BASE and GDD seen in Figure 11, and the smaller differences in concentration between these two simulations seen at Oahu in Figure 12. In the South Pacific, none of the sensitivity studies do a better job at Funafuti, suggesting either that interannual variability is important at this site, or that differences in deposition or transport parameterizations between our simulation and Ginoux *et al.* [2001] are responsible for the differences.

[38] Similar to the optical depth comparisons over Australia, changing either the source parameterization or the meteorology substantially improves the performance at both New Caledonia and Cape Grim in BASE. Figure 12a shows the both GDD and BASE-DD over predict surface dust at Barbados in January and February, while the other cases do not. Since GDD and BASE-DD use different a source parameterization, but the same transport meteorology, this implies that the over predictions seen in January and February (but not November) are caused by transport. At Mace Head in February to April, we see that BASE has the highest over prediction, followed by GDN, BASE-DD and BASE-CULT, while GDD has the smallest over prediction. From this we conclude that high dust at Mace Head in February to April is caused by a combination of source parameterization and meteorology. At Norfolk we see large discrepancies between model and observed monthly mean concentrations. However, Figure 5 shows fairly correct comparison to the data, suggesting that interannual variability at Norfolk is considerable.

[39] Also included in Figure 12 is the case BASE+CULT, which includes a “disturbed soils” source. The dust surface concentration from BASE+CULT simulation are still too high in south hemisphere sites such as New Caledonia, Cape Grim, but are better than the BASE case at these sites. The surface dust concentration calculated by BASE+CULT at Cape Point is much higher than BASE and GDD runs (seen in Figure 12), suggesting that at this site, at least, the crude methodology for calculating “disturbed sources” is not accurate. Except at Cape Point, the BASE+CULT simulation compares similarly to the BASE simulation with the in situ data.

[40] In order to compare daily features of different sensitivity studies, Figure 13 shows the Barbados daily averaged concentrations from observations [Prospero and Nees, 1986; D. Savoie, personal communication, 2001] and the BASE and GDD cases. Both simulations of BASE and GDD capture many events, but also missed some dust events in 1995. From daily comparison, it can be seen that both BASE and GDD over predict dust at Barbados especially in January to February, and July to August and October to November, especially for GDD run, consistent with the monthly mean comparison (seen in Figure 12). The correlation (0.44) between observation and GDD is higher than the correlation (0.33) between observation and BASE,

but this correlation is not statistically significant when we assume that daily averaged concentration are independent after 3 days, which is probably not the case (see Figure 13 or Cakmur and Miller [2001]). The model results are also moderately correlated with each other (0.45).

[41] Overall, the results suggest that the modeling of dust is sensitive to both input meteorological data sets as well as the source parameterization, but different regions have different sensitivities. In addition, these sensitivities are so large that they make distinguishing differences between different sources. From our analysis, it is not possible to say generally speaking which reanalysis is better, although there is strong evidence that the NCEP reanalysis does not do a good job in simulating Southern Hemisphere dust, while the DAO analyses over predicts East Asian dust.

4.3. Vertical Distributions

[42] Finally, we explore differences between modeled vertical profiles within our sensitivity studies. Here we do not have observations to compare against. Figure 14 shows the zonal averaged vertical distribution of dust concentrations in five sensitivity analyses (BASE, GDD, BASE-DD, GDN, and BASE+CULT) for November, 1995. Notice first of all that there is considerably more dust in the Southern Hemisphere in BASE, BASE+CULT than GDD and GDN or BASE-DD, as discussed above. In the case where NCEP winds are used for transport (BASE and GDN, and BASE+CULT), there is a strong gradient in concentration at about 700 mb, while when DAO winds are used for transport, the strong gradient appears closer to 600 mb. If we compare this with the radon vertical profiles over our source areas, we see that radon is transported higher over North Africa and Arabian Peninsular in the NCEP winds compared with the DAO winds, consistent with these zonal averaged dust distributions.

5. Summary and Conclusions

[43] In this paper we present climatological dust distribution and deposition calculated by the MATCH global dust transport model with the Dust Entrainment And Deposition (DEAD) dust module from 1979 to 2000, using NCEP reanalysis data. We compare this climatology to several available observational data sets. The model simulated the main dust sources as those identified by Prospero *et al.* [2002] as the “dust belt”: e.g., North Africa, Arabian Peninsula, East Asia, and North America, with the largest dust concentrations and deposition are concentrated close to the source regions. The pattern of the optical depths is consistent with the AVHRR and MODIS satellite measurements. The comparisons between calculated and AERONET measured optical thickness show that the model simulates the evolution of the monthly average optical depth. The seasonal cycles of surface dust concentrations calculated by the model are similar to the observations at most stations, although the model over predicts surface concentrations at several observing stations in the Southern Hemisphere. The model predicted deposition is similar to in situ measurements over several orders of magnitude.

[44] In our climatological simulation the total emissions of desert dust to the atmosphere are 1654 Tgyr^{-1} on average after being adjusted to match the observations (subjectively),

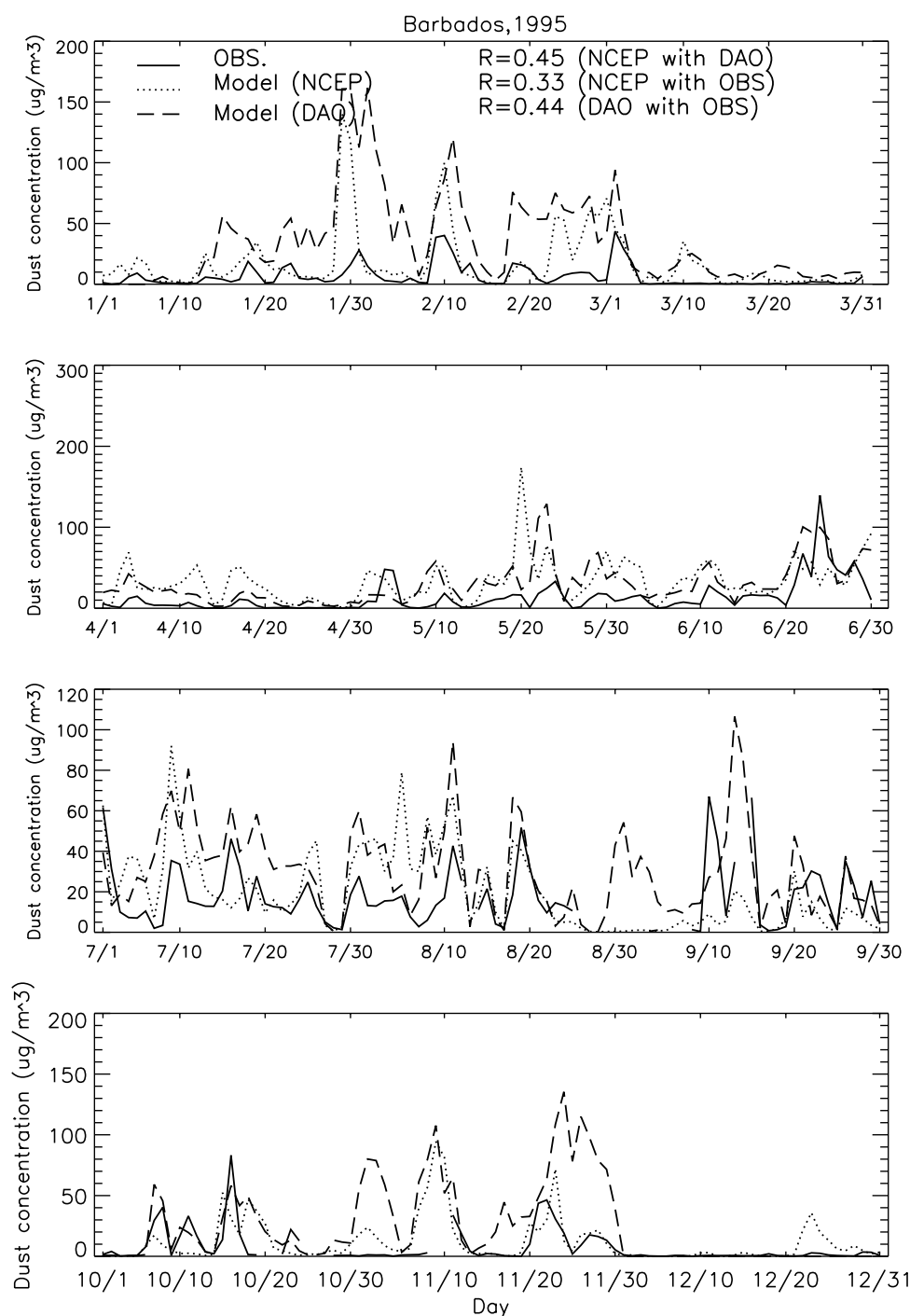


Figure 13. Daily dust concentration comparison between BASE and GDD in 1995, in units of $\mu\text{g m}^{-3}$.

similar to *Ginoux et al.* [2001], and on the high end of the range suggested by *Tegen et al.* [2002]. About a 6-day lifetime of desert dust is similar to previous studies, although the amount of wet deposition is much larger than in *Ginoux et al.* [2001]. In *Ginoux et al.* [2001], they assume a wet deposition lifetime of 56 days, which is longer than most other predictions of aerosol lifetimes [e.g., *Balkanski et al.*, 1993; *Kriedenweis et al.*, 1999; *Guelle et al.*, 2000]. However, these aerosol lifetimes are for all aerosol species, not mineral aerosols, which occur preferentially in arid regions and thus mineral aerosols should have a longer lifetime. The

large discrepancy between the lifetimes predicted in this model, and those used in the *Ginoux et al.* [2001] study, both of which match available observations, suggest that much work is necessary in improving our understanding of mineral aerosol removal by wet deposition. Since wet deposited desert dust particles are thought to contain much more bioavailable iron [e.g., *Jickells and Spokes*, 2001], the partitioning between wet and dry deposition has large biogeochemical implications.

[45] A source apportionment analysis for the year 1998 shows that the African sources are the largest source

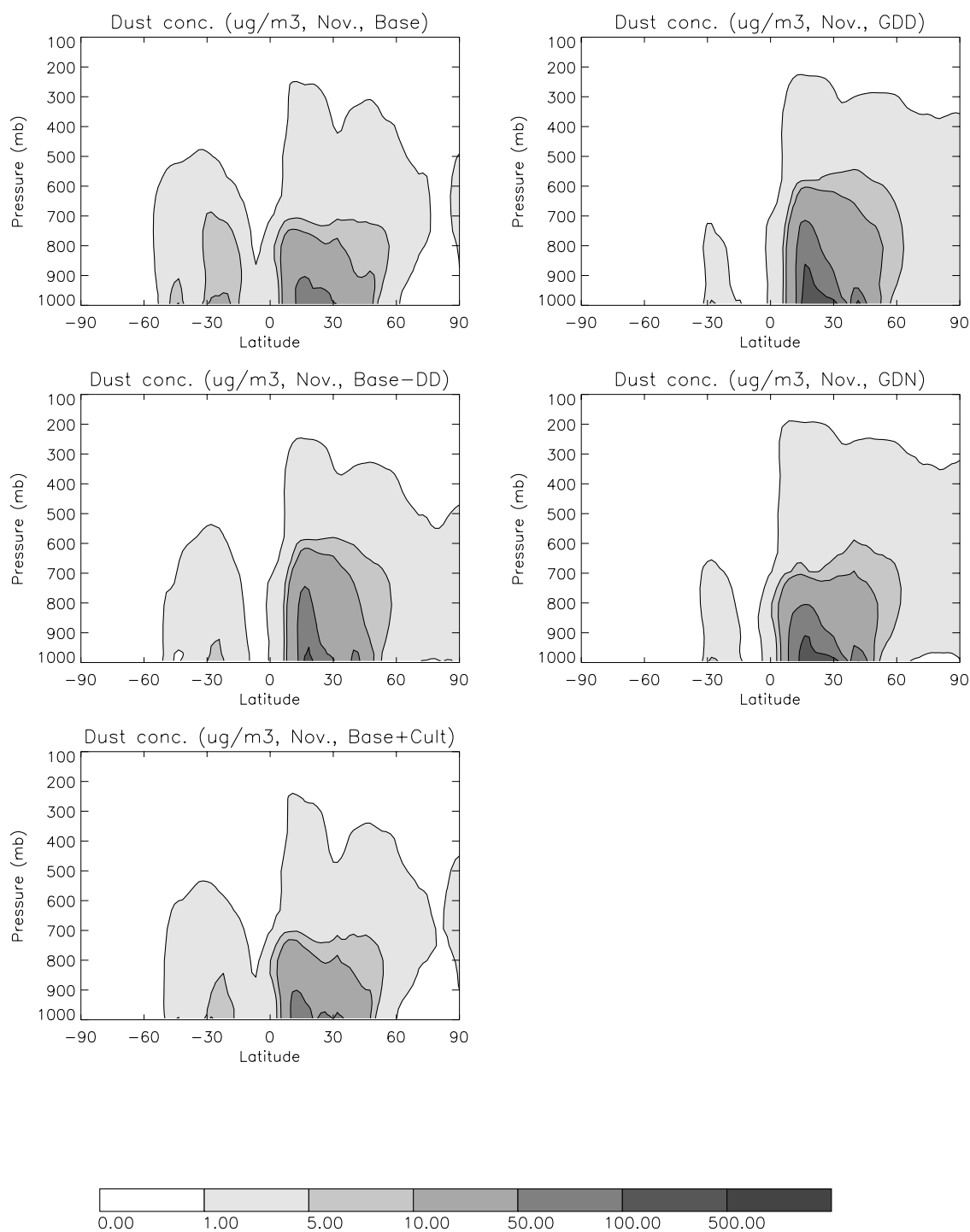


Figure 14. Zonal mean dust concentration from different sensitivity simulations in November 1995, in units of $\mu\text{g m}^{-3}$.

globally in our control run simulation, accounting for 67% of total dust mobilization and 73% of the dust loading. The East Asia source has large contribution to dust distribution in the North Pacific. The Australian source dominated the South Hemisphere dust.

[46] In this paper we have presented several sensitivity analyses of dust simulations results, which include different mobilization, different wind transport, and changes in the processes responsible for source areas. We use these sensi-

tivity studies to understand the differences between the dust climatologies presented here and in *Ginoux et al.* [2001], as well as understand the uncertainties in dust modeling. The main meteorological parameters, such as wind, precipitation, boundary layer height for NCEP/NCAR reanalysis data are consistent with those from the GEOS/DAO reanalysis data, but there are differences, especially in the precipitation patterns. Spatial correlations between precipitation from both NCEP and DAO archived precipitation and the

Xie and Arkin [1996] observational based data set are ~ 0.30 , emphasizing that errors in the hydrological cycle in reanalyses are large, consistent with previous studies [e.g., Trenberth and Guillemot, 1998; Santer et al., 2000].

[47] Comparisons with other modeling studies suggest that the model presented here can simulate the North Pacific dust more accurately than the modeling study of Ginoux et al. [2001], without requiring the inclusion of a phenology, as proposed by Tegen et al. [2002]. Sensitivity studies suggest that the differences between this study and Ginoux et al. [2001] are due to differences in the surface meteorology in East Asia source areas, although we cannot preclude a better cancellation of errors in the BASE case than in Ginoux et al. [2001]. The model simulations presented here, however, do a worse job of matching the observations in the Southern Hemisphere than either Ginoux et al. [2001] or Tegen et al. [2002]. As pointed out by Ginoux et al. [2001], the models are sensitive to the defined preferential source areas, but this study and the comparison with Ginoux et al. [2001] suggests that there is a strong sensitivity to the details of the mobilization and which meteorology data set. There are large differences between the Ginoux et al. [2001] and our climatological simulations in the Southern Hemisphere. Our study suggests that the reasons for these differences are due both to mobilization scheme and the meteorological winds.

[48] We also considered the impact of a source due to cultivation in deserts being responsible for 50% of the atmospheric loading, similar to that hypothesized by Tegen and Fung [1995]. The source areas and mobilization from these sources are quite similar (Figure 1, similar to Mahowald et al. [2002]). The inclusion of a land-use source compared equally well with available observations as the control simulations with only natural sources (except at one site), and the differences between simulations including cultivation as a source and with only topographic lows as sources was similar to the differences in simulations due to changes in meteorological data sets or source parameterizations. Thus, our analysis suggests that the available data sets of desert dust are not able to exclude a land-use source of desert dust due to uncertainties in the meteorology and source parameterizations.

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